

Chapter 1.7 Product Measures, Fubini's Theorem

Rick Durrett Probability: Theory and Examples

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Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.7

Learning goal

Build product measures and use Tonelli–Fubini as the main engine for switching order of integration.

1. Product measures on product sigma-fields.
2. Measurable sections and section functions.
3. Tonelli for nonnegative functions; Fubini for integrable functions.
4. Product measures as the measure-theoretic preview of independence.

Product Measures

Theorem

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be sigma-finite measure spaces. There is a unique measure

$$\mu \times \nu \quad \text{on } \mathcal{A} \otimes \mathcal{B}$$

such that

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B), \quad A \in \mathcal{A}, B \in \mathcal{B}.$$

Pitfall

Sets in $\mathcal{A} \otimes \mathcal{B}$ are usually much more complicated than finite unions of rectangles.

LLMwiki: [Product measures](#); ID: [durrett_ch1_product_measure](#)

Product Measures: Proof Idea

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Measurable Sections

Theorem

If $C \in \mathcal{A} \otimes \mathcal{B}$, define sections

$$C_x = \{y : (x, y) \in C\}, \quad C^y = \{x : (x, y) \in C\}.$$

Then $C_x \in \mathcal{B}$ and $C^y \in \mathcal{A}$. Moreover, if ν is sigma-finite, then

$$x \mapsto \nu(C_x)$$

is $\mathcal{A}$ -measurable.

[LLMwiki: Measurable sections](#); ID: durrett_ch1_sections_measurable

Measurable Sections: How to Prove

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Tonelli

If $f \geq 0$ is measurable on $X \times Y$, then

$$\int f \, d(\mu \times \nu) = \int_X \int_Y f(x, y) \, \nu(dy) \, \mu(dx) = \int_Y \int_X f(x, y) \, \mu(dx) \, \nu(dy),$$

allowing the value $+\infty$.

Fubini

If $\int |f| \, d(\mu \times \nu) < \infty$, the same identities hold with finite signed integrals.

LLMwiki: Fubini and Tonelli theorem; ID: durrett_ch1_fubini_tonelli

Tonelli and Fubini: How to Prove

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Choosing Tonelli or Fubini

Checklist

- If $f \geq 0$, use Tonelli first; infinite answers are allowed.
- If f has signs, prove $\int |f| < \infty$ before switching orders.
- To prove integrability, Tonelli is often applied to $|f|$.
- To compute measures of regions, write the region by sections and integrate their lengths or masses.

LLMwiki: Fubini and Tonelli theorem; ID: durrett_ch1_fubini_tonelli

Preview: Independence

Idea

If a joint law factors as a product measure,

$$\mathcal{L}(X, Y) = \mathcal{L}(X) \times \mathcal{L}(Y),$$

then the two coordinates behave independently.

Pitfall

Independence is a statement about the joint law, not merely about knowing the two marginal laws.

LLMwiki: Product measures as preview of independence; ID: durrett_ch1_independence_preview

Exercise 1.7.1

Show that if

$$\int_X \int_Y |f(x, y)| \mu_2(dy) \mu_1(dx) < \infty,$$

then

$$\int_X \int_Y f(x, y) \mu_2(dy) \mu_1(dx) = \int_{X \times Y} f d(\mu_1 \times \mu_2) = \int_Y \int_X f(x, y) \mu_1(dx) \mu_2(dy).$$

LLM[wiki](#): Fubini and Tonelli theorem; ID: durrett_ch1_fubini_tonelli

Exercise 1.7.1: Proof Skeleton

LLMwiki: Integrable functions; ID: durrett_ch1_integrable_function

Exercise 1.7.2

Let $g \geq 0$ be measurable on $(X, \mathcal{A}, \mu)$. Use Fubini/Tonelli to show

$$\int_X g \, d\mu = (\mu \times \lambda)(\{(x, y) : 0 \leq y < g(x)\}) = \int_0^\infty \mu(\{x : g(x) > y\}) \, dy.$$

LLMwiki: Layer-cake formula via product sets; ID: `exercise_method_layer_cake_formula`

Exercise 1.7.2: Proof Skeleton

LLMwiki: Measurable sections; ID: durrett_ch1_sections_measurable

Exercise 1.7.3

Let F, G be Stieltjes measure functions and let μ, ν be the corresponding measures. Show:

1.

$$\int_{(a,b]} (F(y) - F(a)) dG(y) = (\mu \times \nu)(\{(x, y) : a < x \leq y \leq b\}).$$

2.

$$\int_{(a,b]} F(y) dG(y) + \int_{(a,b]} G(y) dF(y) = F(b)G(b) - F(a)G(a) + \sum_{x \in (a,b]} \mu(\{x\})\nu(\{x\}).$$

3. If $F = G$ is continuous, then

$$\int_{(a,b]} 2F(y) dF(y) = F^2(b) - F^2(a).$$

LLMwiki: Stieltjes integration by parts with jump correction; ID:

`exercise_method_stieltjes_integration_by_parts_with_jumps`

Exercise 1.7.3: Part (i)

LLMwiki: Stieltjes measure; ID: durrett_ch1_stieltjes_measure

Exercise 1.7.3: Parts (ii)–(iii)

LLMwiki: Fubini and Tonelli theorem; ID: durrett_ch1_fubini_tonelli

Exercise 1.7.4

Let μ be a finite measure on $\mathbb{R}$ and

$$F(x) = \mu((-\infty, x]).$$

Show that, for $c > 0$,

$$\int_{\mathbb{R}} (F(x+c) - F(x)) dx = c \mu(\mathbb{R}).$$

LLMwiki: Sliding interval Fubini identity; ID: exercise_method_interval_sliding_fubini

Exercise 1.7.4: Proof Skeleton

LLMwiki: Fubini and Tonelli theorem; ID: durrett_ch1_fubini_tonelli

Exercise 1.7.5

Show that $e^{-xy} \sin x$ is integrable on the strip

$$0 < x < a, \quad 0 < y < \infty.$$

Evaluate the double integral in both orders and obtain the usual sine-integral estimate.

LLMwiki: Sine integral via Fubini; ID: `exercise_method_sine_integral_fubini`

Exercise 1.7.5: Integrability

LLMwiki: Integrable functions; ID: durrett_ch1_integrable_function

Exercise 1.7.5: Two Orders

LLMwiki: Fubini and Tonelli theorem; ID: durrett_ch1_fubini_tonelli

Exercise 1.7.5: Estimate

LLMwiki: Sine integral via Fubini; ID: `exercise_method_sine_integral_fubini`

Chapter 1.7 Summary

- Product measures are determined by measurable rectangles.
- Section arguments reduce product-space questions to one-coordinate questions.
- Tonelli handles nonnegative functions, even with infinite values.
- Fubini handles signed functions after absolute integrability is known.
- Many probability identities are just product-measure counts in disguise.

LLMwiki: [Product measures](#); ID: `durrett_ch1_product_measure`